

HISABDO AI

DAY 10 — FINANCIAL HEALTH SCORE

Professional MVP Specification & Implementation Guide

1. Document Purpose

This document defines the Day 10 Financial Health Score module for HisabDo AI. The objective is to establish a clear, explainable and implementable scoring model that converts a user's financial behavior into a score from 0 to 100 and provides an understandable explanation of the result.

2. Business Objective

HisabDo AI should move beyond recording transactions. The Financial Health Score gives the user a quick view of their financial condition and identifies areas that are strong or need attention. Example: Financial Health Score = 80/100 — Good.

3. Scope of Day 10

Define the financial factors, weights, formulas, thresholds, overall score categories, required HisabDo data, backend calculation flow, AI explanation format and API output required by mobile and web applications.

4. Financial Health Score Model

The initial MVP model contains six factors. The weights total 100 points.

Factor	Weight	Definition
Saving Behavior	25 points	Measures how much income remains after expenses and the user's saving rate.
Expense Control	20 points	Measures expenses relative to income.
Budget Control	20 points	Measures how closely actual spending follows the user's planned budget.
Cash Flow	15 points	Measures the difference between money coming in and money going out.
Debt / Udhaar	10 points	Measures the user's outstanding debt/udhaar burden and related behavior.
Expense Growth	10 points	Measures whether spending is increasing or decreasing over comparable periods.

5. Factor — Saving Behavior

Saving Behavior measures how much of the user's income remains after expenses. A higher saving rate receives a higher score.

Formula: Saving Rate = $((\text{Income} - \text{Expense}) / \text{Income}) \times 100$

Condition	Score
25% or more	25/25
20–24.99%	22/25
15–19.99%	18/25
10–14.99%	14/25
5–9.99%	8/25
Below 5%	3/25

Note: Example: Income PKR 150,000 and Expense PKR 108,500 gives Saving PKR 41,500 and Saving Rate 27.67%.

6. Factor — Expense Control

Expense Control measures how much of income is consumed by expenses. Lower expense ratio generally indicates stronger control.

Formula: Expense Ratio = $(\text{Expense} / \text{Income}) \times 100$

Condition	Score
50% or less	20/20
51–60%	18/20
61–70%	15/20
71–80%	12/20
81–90%	7/20
Above 90%	3/20

7. Factor — Budget Control

Budget Control compares actual expenses with the user's planned budget. Spending within or close to budget receives a higher score.

Formula: Budget Utilization = $(\text{Actual Expense} / \text{Budget}) \times 100$

Condition	Score
100% or less	20/20
101–105%	17/20
106–115%	13/20
116–125%	8/20
Above 125%	3/20

8. Factor — Cash Flow

Cash Flow measures the difference between income and expenses for the analysis period. Positive cash flow is preferred.

Formula: Cash Flow = Total Income – Total Expense

Condition	Score
Strong positive	15/15
Positive	12/15
Near zero	7/15
Negative	3/15

Note: Future iterations can include recurring expenses and forecasted cash flow.

9. Factor — Debt / Udhaar

This factor evaluates outstanding debt/udhaar information available in HisabDo. Lower financial burden and healthier repayment/receivable behavior should receive a higher score.

Formula: Initial MVP ratio/threshold model to be calibrated using actual HisabDo debt data.

Condition	Score
None or very low	10/10
Low	8/10
Moderate	6/10
High	3/10
Very high	0–2/10

Note: Final thresholds must be validated after the actual HisabDo debt/udhaar schema and real sample data are available.

10. Factor — Expense Growth

Expense Growth compares current spending with the previous comparable period. Rapid increases can reduce financial health.

Formula: Growth % = ((Current Expense – Previous Expense) / Previous Expense) × 100

Condition	Score
Decreasing	10/10
0–5% increase	8/10
5–10% increase	6/10
10–20% increase	3/10
Above 20% increase	1/10

11. Overall Score

The final score is the sum of the six factor scores. Maximum = 100 points.

- Saving Behavior = 22/25
- Expense Control = 15/20
- Budget Control = 17/20
- Cash Flow = 12/15
- Debt/Udhaar = 8/10
- Expense Growth = 6/10
- Final Financial Health Score = 80/100

12. Score Classification

Score	Status
90–100	Excellent
75–89	Good
60–74	Fair
40–59	Needs Attention
0–39	Poor

13. AI Explanation Layer

The LLM explains verified financial results in natural language. It must not invent or independently determine financial numbers.

Verified backend values may include:

- Final score
- Score status
- Saving rate
- Expense ratio
- Budget status
- Cash-flow status
- Expense growth
- Top contributing issues

Example: “Your financial health is good with a score of 80/100. Your saving rate and positive cash flow are strong. However, your expenses increased compared with the previous period, so controlling higher-spending categories could improve your score.”

14. Golden Rule — No AI-Generated Financial Numbers

All financial totals, percentages, scores and forecast values must originate from trusted HisabDo data and a deterministic calculation/analytics layer. The LLM receives verified values and converts them into a clear explanation.

Correct flow: HisabDo Database → Calculation/Analytics Engine → Verified Results → LLM Explanation → User

15. Required HisabDo Data

Data	Purpose	Priority
User ID	Keep calculations isolated to the correct user.	Required
Transaction ID	Identify each financial transaction.	Required
Transaction Type	Separate income and expense.	Required
Amount	Perform financial calculations.	Required
Category	Analyze spending behavior.	Required
Date	Compare periods and detect trends.	Required
Budget	Measure budget control.	Required for budget score
Recurring Expense	Improve cash-flow analysis.	Recommended
Customer / Person	Support udhaar and natural-language queries.	Recommended
Debt / Udhaar	Evaluate debt/receivable behavior.	Recommended

16. Backend Processing Flow

1) Authenticate user → 2) Retrieve only that user's permitted records → 3) Aggregate income and expenses → 4) Calculate each factor → 5) Add weighted scores → 6) Assign status → 7) Generate structured insights → 8) Optionally send verified values to LLM → 9) Return API response.

17. Proposed API Contract

Endpoint: GET /api/ai/financial-health/{userId}

Example response:

```
{
  "score": 80,
  "status": "Good",
  "savingRate": 27.67,
  "expenseRatio": 72.33,
  "budgetStatus": "Within Budget",
  "cashFlow": 41500,
  "expenseGrowth": 18,
  "insights": [
    "Saving behavior is healthy",
    "Expenses increased compared with the previous period"
  ]
}
```

18. Mobile & Web Output

The same backend result should be reusable by Flutter mobile and web applications. The UI should show the score, status, factor-level strengths/weaknesses and an AI-generated explanation.

- Financial Health
- 80/100
- Good
- Saving — Strong
- Expense Control — Good
- Budget — Good
- Cash Flow — Positive
- Debt/Udhaar — Healthy
- Expense Growth — Needs Attention
- AI Insight

19. Testing Requirements

- Verify income, expense and saving calculations manually.
- Verify every scoring threshold.
- Test zero-income users.
- Test users with no transactions.
- Test users with no budget.
- Test negative cash flow.

- Test very large transactions.
- Test missing/invalid categories.
- Test duplicate transactions.
- Test user-data isolation and authorization.
- Verify that LLM cannot override verified numbers.

20. Day 10 Team Deliverables

- ☐ Approved scoring model
- ☐ Factor formulas and thresholds
- ☐ Required data specification
- ☐ Backend calculation design
- ☐ API contract
- ☐ AI explanation specification
- ☐ Mobile/Web UI design
- ☐ Test cases

21. Team Lead Approval Checklist

- ☐ Every factor and its purpose are understood.
- ☐ Total weights equal 100 points.
- ☐ All formulas are documented.
- ☐ Financial calculations are outside the LLM.
- ☐ Required HisabDo data is identified.
- ☐ API input/output is defined.
- ☐ Edge cases are listed.
- ☐ Model is clearly marked as MVP and will be validated against real data.

22. Final Definition

HisabDo Financial Health Score is an explainable 0–100 financial wellness indicator that evaluates saving, expense control, budget adherence, cash flow, debt/udhaar and expense growth using verified HisabDo data. The calculation engine determines the score, while AI explains the result and highlights areas that may require attention.

DAY 10 OUTPUT — READY FOR TEAM IMPLEMENTATION